

## Orthopedic Driven Imaging

# AP/PA Annotation Definitions

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**Version:** 1.0.0

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**Effective Date:** \_\_\_\_\_

**Approved By:** \_\_\_\_\_



Company: Orthopedic Driven Imaging, LLC

## Contents

|          |                                                |           |
|----------|------------------------------------------------|-----------|
| <b>1</b> | <b>Preamble</b>                                | <b>2</b>  |
| <b>2</b> | <b>Medical Approval</b>                        | <b>4</b>  |
| <b>3</b> | <b>Image Annotations</b>                       | <b>5</b>  |
| 3.1      | Ilium Contour                                  | 5         |
| 3.2      | Pelvic Outlet Contour                          | 6         |
| 3.3      | Pelvic Teardrops                               | 7         |
| 3.4      | Ischial Tuberosities                           | 8         |
| 3.5      | Pubic Symphysis Contour                        | 9         |
| 3.6      | Distal Sacroiliac Joint                        | 10        |
| 3.7      | Acetabular Contour                             | 11        |
| 3.8      | Femoral Head Contour                           | 12        |
| 3.9      | Greater Trochanter Contour                     | 13        |
| 3.10     | Proximal Medial Femur Contour                  | 14        |
| 3.11     | Lesser Trochanter Contour                      | 15        |
| 3.12     | Distal Medial Femur Contour                    | 16        |
| 3.13     | Obturator Foramen Contour                      | 18        |
| <b>4</b> | <b>Annotation Order</b>                        | <b>19</b> |
| <b>5</b> | <b>General Guidance for Contour Annotation</b> | <b>21</b> |
| 5.1      | What Is a Contour?                             | 21        |
| 5.2      | How to Annotate a Contour                      | 21        |
| 5.3      | Partly Obscured or Off-Screen Contours         | 21        |
| <b>6</b> | <b>Derived Features of Interest</b>            | <b>23</b> |
| <b>7</b> | <b>Categorical Annotation</b>                  | <b>33</b> |
| <b>A</b> | <b>Annotation ID Summary Table</b>             | <b>34</b> |

## 1. Preamble

This document defines annotations (i.e., image features such as points, lines and edge contours) within anteroposterior (AP) or posteroanterior (PA) radiographic/fluoroscopic images. It is intended to provide a consistent reference for annotation placement within the annotation tool in order to define a dataset for downstream machine learning tasks.

Each annotation definition includes:

- the identifying annotation IDs;
- a definition of the annotation;
- optional further notes relevant to interpretation or repeatability; and
- visual examples.

The document includes the following sections:

- **Medical Approval:** Records review of the document for medical appropriateness and clinical usability.
- **Image Annotations:** Defines the points, lines and contours to be placed directly by annotators.
- **Annotation Order:** Specifies the recommended sequence for completing annotations so that prerequisite anatomical references are established before dependent annotations are placed.
- **General Guidance for Contour Annotation:** Explains what a contour is, how to trace a visible boundary and how to handle contours that are partly obscured or extend beyond the image.
- **Derived Features of Interest:** Defines points, lines and other measurements that are deterministically calculated from one or more source annotations and subsequently presented to the annotator for approval.
- **Categorical Annotation:** Defines the image-level categorical classifications that annotators are required to assign.
- **Annotation ID Summary Table:** Provides a consolidated reference for the annotation identifiers used throughout the document.

Derived features are not placed independently by the annotator. They are calculated automatically from the accepted source annotations according to the definitions provided in this document. Annotators are responsible for reviewing the generated features and categorising each as correct, incorrect or missing, as applicable.

Unless otherwise stated, left- and right-sided annotations use equivalent definitions reflected across the image. References to superior, inferior, medial, lateral, proximal and distal directions are anatomical references and should be interpreted with respect to the patient rather than the displayed orientation of the image.

All annotations must be selected according to the visible features present in the image under review. For directly traced contours, annotators must stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across any obscured region. Annotators should not infer the position of an anatomical boundary through a region in which that boundary is not visible unless the applicable annotation definition explicitly instructs them to fit or approximate a geometric structure.

When an annotation is visible but its exact placement cannot be determined consistently, the annotator should flag it for further review and may include a note describing the source of uncertainty. An annotation which is clearly not visible, not present or not applicable should instead be recorded using the corresponding missing or non-applicable status available within the annotation tool.

Where the annotation tool automatically constrains, fits or derives a feature, the resulting feature must still satisfy the anatomical and geometric definition given in this document. Tool-generated placement does not replace annotator review or the approval criteria specified for the feature.

## 2. Medical Approval

This annotation definition document has been reviewed for medical appropriateness and clinical usability.

**Approved by:**

**Surgeon Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_

**Signature:** \_\_\_\_\_ **Institution / Department:** \_\_\_\_\_

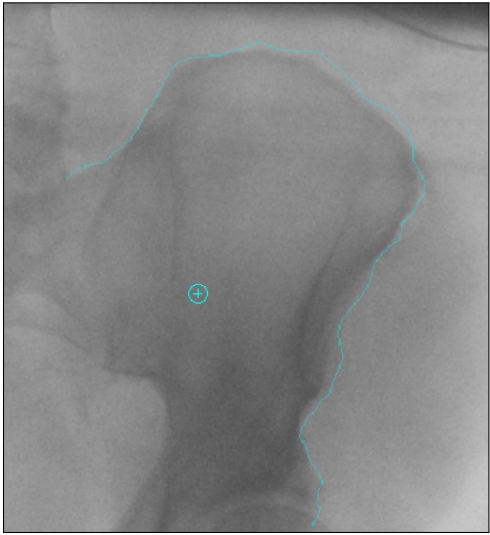
**Comments / approval notes:**

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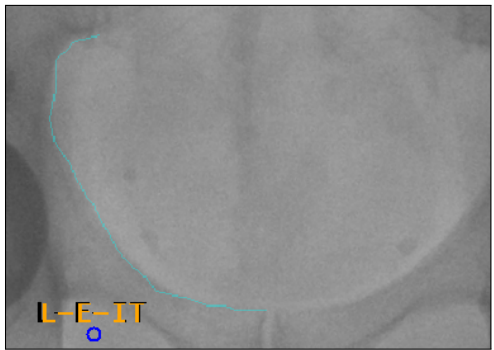
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### 3. Image Annotations

#### 3.1. Ilium Contour

| Ilium Contour     |                                                                                                                                                                                                                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs    | L-C-ILI Left Superior-Lateral Ilium Contour                                                                                                                                                                                                                                                                                                  |
|                   | R-C-ILI Right Superior-Lateral Ilium Contour                                                                                                                                                                                                                                                                                                 |
| Definition        | The contour that traces from the edge of the spine along the sacrum and ilium down to the acetabular rim. Although the annotation is named after the ilium, portions of the contour mark the superior boundary of the sacrum.                                                                                                                |
| Further Notes     | Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary. |
| Visual Example(s) | <div data-bbox="719 913 1206 1444"></div> <p>An example of a left ilium contour.</p>                                                                                                                                                                      |

### 3.2. Pelvic Outlet Contour

| Pelvic Outlet Contour |                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs        | L-C-P0 Left Pelvic Outlet Contour                                                                                                                                                                                                                                                                                                            |
|                       | R-C-P0 Right Pelvic Outlet Contour                                                                                                                                                                                                                                                                                                           |
| Definition            | The contour that traces the boundary of the pelvic outlet from the superior margin of the pubic symphysis to the distal sacroiliac joint.                                                                                                                                                                                                    |
| Further Notes         | Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary. |
| Visual Example(s)     | <div data-bbox="717 751 1208 1098"></div> <p>An example of a right pelvic outlet contour.</p>                                                                                                                                                             |

### 3.3. Pelvic Teardrops

#### Pelvic Teardrops

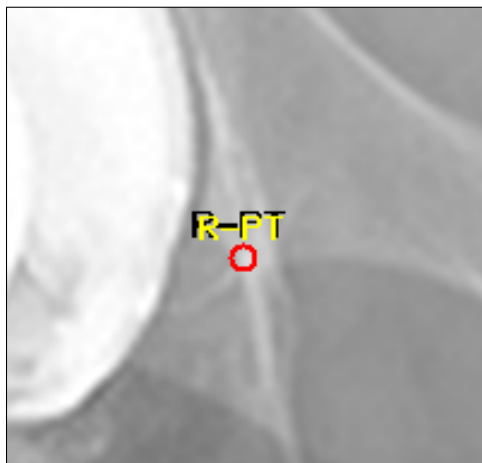
##### Annotation IDs

|      |                       |
|------|-----------------------|
| L-PT | Left Pelvic Teardrop  |
| R-PT | Right Pelvic Teardrop |

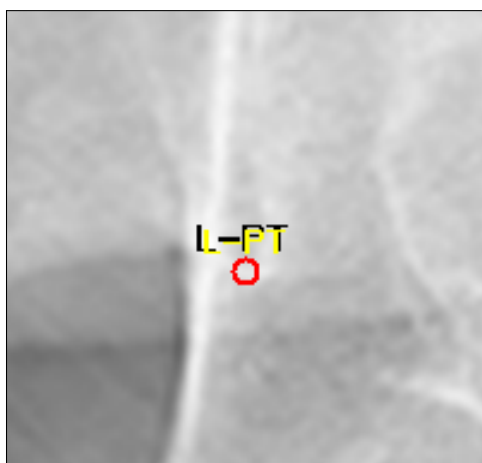
##### Definition

Select the most inferior edge at the bottom of the teardrop.

##### Visual Example(s)

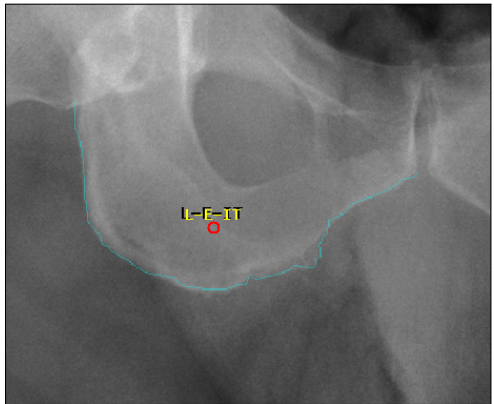


An example of a right pelvic teardrop landmark near an implant.

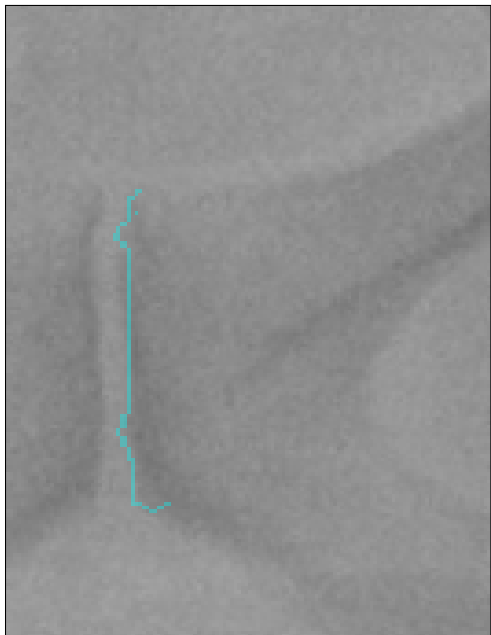


An example of a left pelvic teardrop landmark.

### 3.4. Ischial Tuberosities

| Ischial Tuberosities |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs       | L-C-IT Left Ischial Tuberosity Contour                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                      | R-C-IT Right Ischial Tuberosity Contour                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Definition           | The contour that traces the inferior boundary of the ischial tuberosity from the inferior margin of the pubic symphysis to the point at which the lateral ischium meets the acetabular rim. If the femur obscures this boundary before that endpoint is reached, stop at the last clearly visible point. The contour must not trace along an ischial boundary that overlaps with the femur.                                                                    |
| Further Notes        | The point at which the lateral ischium meets the acetabular rim is the superior-lateral endpoint of this contour. Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary. |
| Visual Example(s)    |  <p>An example of a right ischial tuberosity contour.</p>                                                                                                                                                                                                                                                                                                                   |

### 3.5. Pubic Symphysis Contour

| Pubic Symphysis Contour |                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs          | L-C-PS Left Pubic Symphysis Contour                                                                                                                                                                                                                                                                                                          |
|                         | R-C-PS Right Pubic Symphysis Contour                                                                                                                                                                                                                                                                                                         |
| Definition              | The contour that spans one side of the pubic symphysis from its superior margin to its inferior margin; that is, between the apparent corner points formed by the superior and inferior margins of the pubis along the border of the pubic symphysis. If both internal and external contours are visible, choose the most medial contour.    |
| Further Notes           | Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary. |
| Visual Example(s)       |  <p>An example of a right pubic symphysis contour.</p>                                                                                                                                                                                                    |

### 3.6. Distal Sacroiliac Joint

#### Distal Sacroiliac Joint

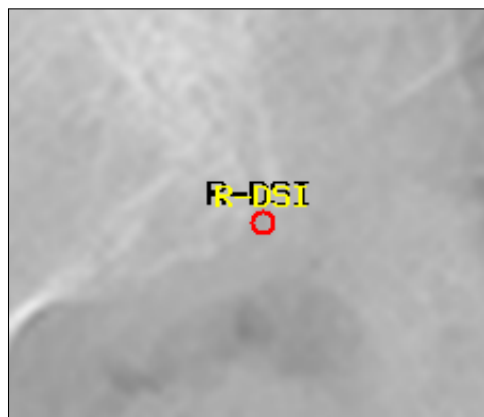
##### Annotation IDs

|       |                               |
|-------|-------------------------------|
| L-DSI | Left Distal Sacroiliac Joint  |
| R-DSI | Right Distal Sacroiliac Joint |

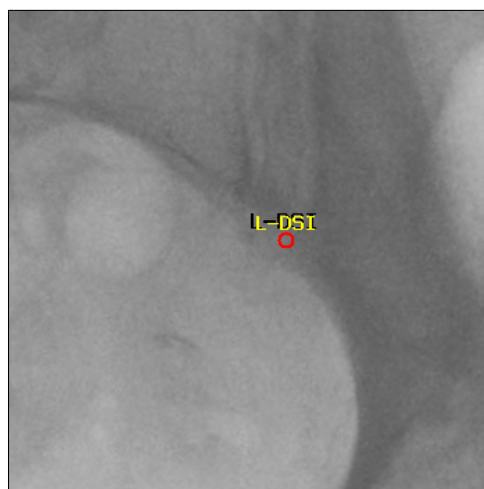
##### Definition

Select the point at the visible distal termination of the sacroiliac (SI) joint between the visible bone edges (centre of the gap). If multiple joint lines are visible, choose the most lateral.

##### Visual Example(s)

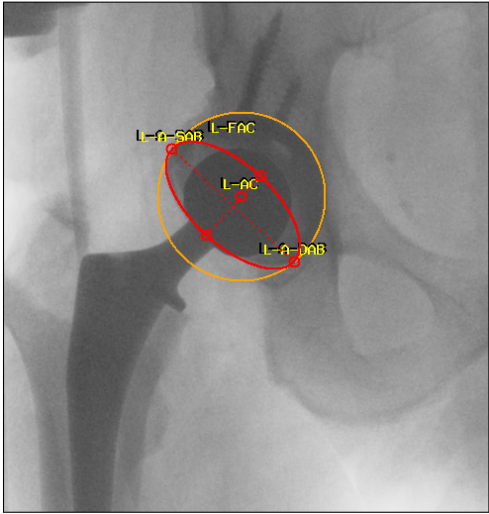


An example of a right distal sacroiliac joint landmark.

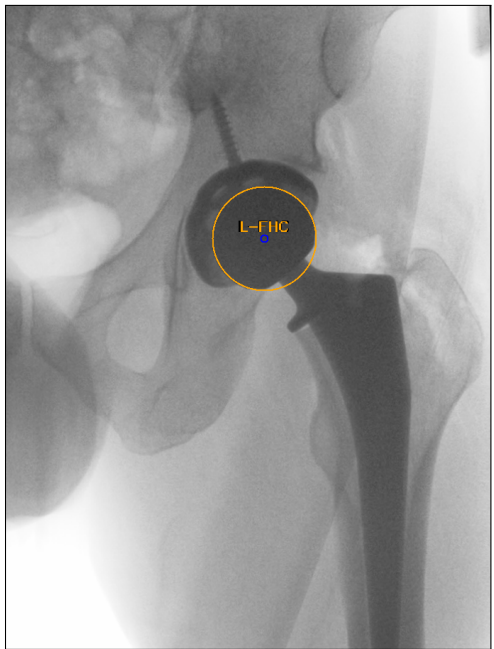


An example of a correctly selected left distal sacroiliac joint landmark in the presence of two apparent joint lines.

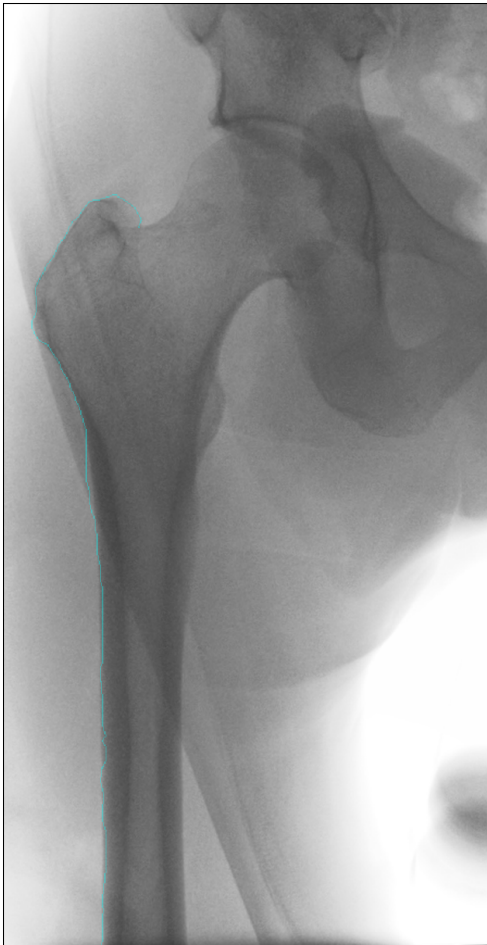
3.7. Acetabular Contour

| Acetabular Contour |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs     | L-C-AC Left Acetabular Contour                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                    | R-C-AC Right Acetabular Contour                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                    | L-C-FAC Left Acetabular Face Contour                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                    | R-C-FAC Right Acetabular Face Contour                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Definition         | For an acetabular implant, define both the spherical contour of the cup and the elliptical contour of its face. Place the spherical contour first using the <b>hover circle tool</b> , fitting it to the spherical portion of the acetabular cup implant. Then place the elliptical contour using the <b>acetabular face tool</b> , which snaps the major-axis endpoints to the fitted circle. Adjust the inclination by dragging the major-axis endpoints around the circle, and expand or shrink the minor axis to match the visible contour of the implant face, typically around its inferior portion. Do not annotate either the acetabular spherical contour or the acetabular face contour for a native hip. |
| Further Notes      | Apply the geometric fit only when enough of the relevant boundary is visible to position it reliably. If the available boundary is insufficient, record the annotation as missing rather than estimating its placement from unrelated anatomy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Visual Example(s)  | <div></div> <p>An example of a right acetabular spherical contour and elliptical face contour.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

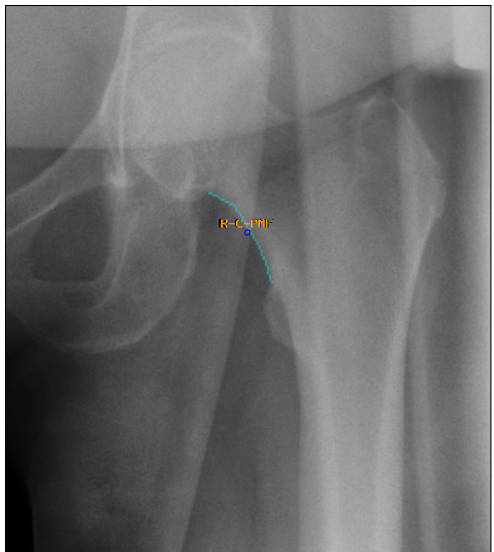
### 3.8. Femoral Head Contour

| Femoral Head Contour |                                                                                                                                                                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs       | L-C-FHC Left Femoral Head Contour                                                                                                                                                                                                              |
|                      | R-C-FHC Right Femoral Head Contour                                                                                                                                                                                                             |
| Definition           | For a native hip, use the hover circle to approximate the contour of the visible bony femoral head. For a prosthetic hip, fit the hover circle to the outer margin of the femoral head implant.                                                |
| Further Notes        | Apply the geometric fit only when enough of the relevant boundary is visible to position it reliably. If the available boundary is insufficient, record the annotation as missing rather than estimating its placement from unrelated anatomy. |
| Visual Example(s)    | <div data-bbox="717 751 1208 1396"></div> <p>An example of a left femoral head contour on an implant.</p>                                                   |

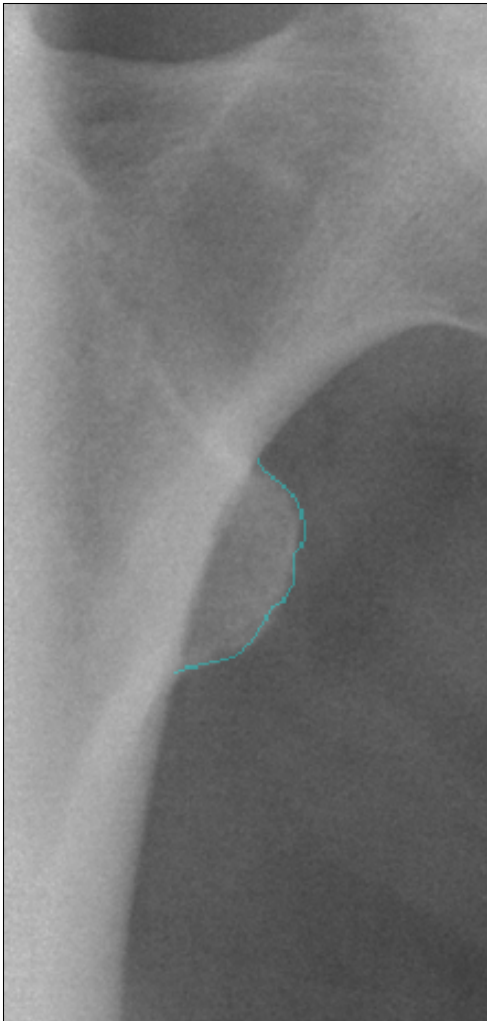
### 3.9. Greater Trochanter Contour

| Greater Trochanter Contour |                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs             | L-C-GT Left Greater Trochanter Contour                                                                                                                                                                                                                                                                                                       |
|                            | R-C-GT Right Greater Trochanter Contour                                                                                                                                                                                                                                                                                                      |
| Definition                 | The contour along the superior-lateral boundary of the femur from the neck-greater trochanter transition point to the most distal visible lateral point in the image.                                                                                                                                                                        |
| Further Notes              | Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary. |
| Visual Example(s)          |  <p>An example of a right greater trochanter contour.</p>                                                                                                                                                                                                 |

### 3.10. Proximal Medial Femur Contour

| Proximal Medial Femur Contour |                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs                | L-C-PMF Left Proximal Medial Femur Contour                                                                                                                                                                                                                                                                                                   |
|                               | R-C-PMF Right Proximal Medial Femur Contour                                                                                                                                                                                                                                                                                                  |
| Definition                    | The contour along the medial boundary of the proximal femur from the proximal tip of the lesser trochanter boundary up the neck to the femoral head.                                                                                                                                                                                         |
| Further Notes                 | Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary. |
| Visual Example(s)             | <div data-bbox="717 787 1208 1335"></div> <p>An example of a left proximal medial femur contour.</p>                                                                                                                                                      |

### 3.11. Lesser Trochanter Contour

| Lesser Trochanter Contour                        |                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs                                   | L-C-LT Left Lesser Trochanter Contour                                                                                                                                                                                                                                                                                                        |
|                                                  | R-C-LT Right Lesser Trochanter Contour                                                                                                                                                                                                                                                                                                       |
| Definition                                       | The contour along the visible medial boundary of the lesser trochanter, beginning at its proximal transition from the medial femur and ending at its distal transition back to the medial femoral shaft.                                                                                                                                     |
| Further Notes                                    | Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary. |
| Visual Example(s)                                |                                                                                                                                                                                                                                                           |
| An example of a right lesser trochanter contour. |                                                                                                                                                                                                                                                                                                                                              |

### 3.12. Distal Medial Femur Contour

| Distal Medial Femur Contour |                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs              | L-C-DMF Left Distal Medial Femur Contour                                                                                                                                                                                                                                                                                                     |
|                             | R-C-DMF Right Distal Medial Femur Contour                                                                                                                                                                                                                                                                                                    |
| Definition                  | The contour along the medial boundary of the femur from the distal tip of the lesser trochanter to the most distal visible medial point in the image.                                                                                                                                                                                        |
| Further Notes               | Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary. |

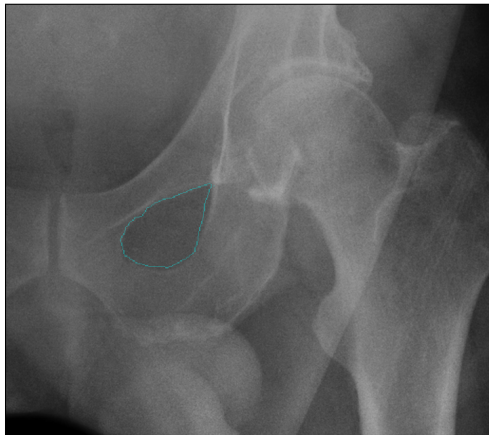
## Distal Medial Femur Contour (Continued)

### Visual Example(s)



An example of a left distal medial femur contour.

### 3.13. Obturator Foramen Contour

| Obturator Foramen Contour |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotation IDs            | L-C-OF Left Obturator Foramen Contour                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                           | R-C-OF Right Obturator Foramen Contour                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Definition                | The contour along the visible boundary of the obturator foramen.                                                                                                                                                                                                                                                                                                                                                                                                              |
| Further Notes             | Annotate only the visible portion of the boundary. If the contour is obscured, stop at the last clearly visible point and restart when the boundary becomes visible again, leaving a gap across the obscured region. Do not visually interpolate across an occlusion. If the contour continues beyond the image, stop at the image boundary.                                                                                                                                  |
| Visual Example(s)         | <div data-bbox="717 716 1203 1148">An anteroposterior radiograph of a human pelvis. The left obturator foramen is clearly visible and its boundary is highlighted with a red contour line. The surrounding bony structures, including the ilium, ischium, and pubis, are visible in grayscale.</div> <p data-bbox="699 1171 1221 1199">An example of a left obturator foramen contour.</p> |

## 4. Annotation Order

To support consistent annotation placement, the image annotations should be completed in the order listed below. The sequence is shown for the left side only; the equivalent right-sided image annotations should be completed in the same relative order.

The order is intended to establish clear anatomical reference points before annotations which depend on those points, to complete related pelvic structures together, and to leave trickier annotations until the surrounding anatomy has been defined.

1. **Left Ilium Contour** ( **L-C-ILI** )

Annotate the ilium contour first to establish the broad superior and lateral boundary of the hemipelvis.

2. **Left Pubic Symphysis Contour** ( **L-C-PS** )

Annotate the pubic symphysis contour before the pelvic outlet and ischial tuberosity contours because its superior and inferior margins define the starting points of those annotations.

3. **Left Distal Sacroiliac Joint** ( **L-DSI** )

Select the distal termination of the sacroiliac joint before drawing the pelvic outlet contour. This point defines the lateral endpoint of the pelvic outlet contour and reduces variation in where that contour is terminated.

4. **Left Pelvic Outlet Contour** ( **L-C-PO** )

Draw the pelvic outlet contour after the pubic symphysis and distal sacroiliac joint have been identified, as these annotations define its endpoints.

5. **Left Obturator Foramen Contour** ( **L-C-OF** )

Annotate the obturator foramen after the adjacent pubic symphysis and pelvic outlet boundaries have been established.

6. **Left Ischial Tuberosity Contour** ( **L-C-IT** )

Annotate the ischial tuberosity contour after the inferior margin of the pubic symphysis and the surrounding pelvic anatomy have been defined.

7. **Left Pelvic Teardrop** ( **L-PT** )

Select the pelvic teardrop after the surrounding pelvic contours have been annotated.

8. **Left Acetabular Contour** ( **L-C-AC** )

For a prosthetic hip, fit the spherical acetabular contour before placing the acetabular face contour. The spherical contour establishes the underlying cup geometry and provides the reference circle to which the acetabular face tool is constrained.

Native hips should not be annotated with an acetabular contour.

9. **Left Acetabular Face Contour** ( **L-C-FAC** )

Place the acetabular face contour immediately after the spherical acetabular contour. The major-axis endpoints are snapped to the spherical contour, so the spherical contour must already be correctly positioned before the face ellipse is adjusted.

Native hips should not be annotated with an acetabular face contour.

10. **Left Femoral Head Contour** ( **L-C-FHC** )

Fit the femoral head contour after completing the acetabular annotations.

11. **Left Greater Trochanter Contour** ( **L-C-GT** )

Annotate the greater trochanter contour after the femoral head has been defined.

12. **Left Proximal Medial Femur Contour** ( **L-C-PMF** )

Trace the proximal medial femoral boundary before annotating the lesser trochanter. Stop at the proximal tip of the lesser trochanter, where the lesser trochanter contour begins.

13. **Left Lesser Trochanter Contour** ( **L-C-LT** )

Annotate the lesser trochanter contour after the proximal medial femur contour has been completed. Trace only the visible medial boundary of the lesser trochanter, from its proximal transition to its distal transition.

14. **Left Distal Medial Femur Contour** ( **L-C-DMF** )

Begin the distal medial femur contour at the distal tip of the lesser trochanter and continue along the medial femoral boundary. The lesser trochanter contour should be completed first so that the transition between the two annotations is consistently identified.

15. **Left Femoral Axis** ( **L-FA** )

Annotate the femoral axis last.

## General Ordering Principles

- Complete all image annotations for one side before beginning the opposite side.
- Establish reference points before drawing contours which begin or end at those points.
- Place the spherical acetabular contour before the acetabular face contour because the face tool depends on the fitted sphere.
- Complete proximal femoral annotations before distal shaft annotations.
- Skip annotations which are not applicable, such as acetabular implant contours on a native hip.

## 5. General Guidance for Contour Annotation

### 5.1. What Is a Contour?

A **contour** describes the visible outer boundary of an anatomical structure, implant or instrument in the two-dimensional image. For model-training purposes, a continuous contour is converted into a discrete set of pixels at the resolution supplied to neural networks. A pixel belongs to the discrete contour set when the continuous contour intersects the area assigned to that pixel. For boundary cases, all four edges of a pixel are considered as belonging to that pixel (for example, a perfectly straight line on the boundary between two pixels would count both of those pixels as part of the discrete contour set).

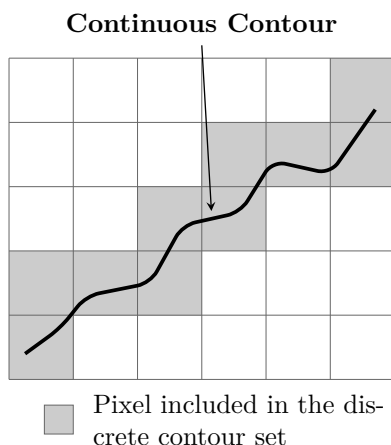


Figure 1: Illustration of discrete contour membership.

### 5.2. How to Annotate a Contour

- Follow the visible edge of the structure as closely as possible.
- The annotation tool will automatically highlight regions at the level of resolution supplied to the neural networks.
- Do not trace the edge of an overlapping implant, instrument, soft-tissue structure or different bone unless the definition explicitly instructs you to do so.

### 5.3. Partly Obscured or Off-Screen Contours

Annotate only the portions of the boundary that are clearly visible. If the contour is hidden by an implant, surgical instrument, table hardware, another anatomical structure or any other object, stop at the last clearly visible point and restart when the same boundary becomes visible again. Leave the hidden section blank rather than visually interpolating across it. A single anatomical contour may therefore contain two or more separate annotated segments.

If the contour continues beyond the edge of the image, stop at the image boundary. Do not estimate the missing portion. If the missing section prevents a downstream feature from being

calculated correctly, classify that downstream feature as *incorrect* or *missing*, as appropriate, during categorical review.

## 6. Derived Features of Interest

Many of the contour annotations have associated features of interest which are deterministically calculated from the contour set. These will be auto-generated by the annotation tool and annotators should ensure that these downstream features meet the terms of their definitions (defined in the tables below).

| Contour ID | Source Contour                 | No. | Derived Feature ID / Name       | Definition and Approval Criteria                                                                                                                                                                                                                                       |
|------------|--------------------------------|-----|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-ILI      | Superior-Lateral Ilium Contour | 1   | SIP<br>Superior Ilium Point     | The most superior point on the ilium contour. If more than one contour point has the same superior-most vertical position, select the most lateral of the tied points. The principal downstream value of interest is the vertical coordinate of the selected point.    |
|            |                                | 2   | LIP<br>Lateral Innominate Point | The most lateral point on the ilium contour. If more than one contour point has the same lateral-most horizontal position, select the most superior of the tied points. The principal downstream value of interest is the horizontal coordinate of the selected point. |

| Contour ID | Source Contour        | No. | Derived Feature ID / Name          | Definition and Approval Criteria                                                                                                                                                                                                                                                                          |
|------------|-----------------------|-----|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-PO       | Pelvic Outlet Contour | 1   | LP0<br>Lateral Pelvic Outlet Point | The most lateral point on the pelvic outlet contour. If more than one contour point has the same lateral-most horizontal position, select the point at the median vertical position among the tied points. The principal downstream value of interest is the horizontal coordinate of the selected point. |
|            |                       | 2   | POD<br>Pelvic Outlet Diameter      | The line segment between the two lateral pelvic outlet points.                                                                                                                                                                                                                                            |

| Contour ID | Source Contour             | No. | Derived Feature ID / Name               | Definition and Approval Criteria                                                                                                                                                                                                                                                 |
|------------|----------------------------|-----|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-IT       | Ischial Tuberosity Contour | 1   | IT<br>Inferior Ischial Tuberosity Point | The most inferior point on the ischial tuberosity contour. If more than one contour point has the same inferior-most vertical position, select the most lateral of the tied points. The principal downstream value of interest is the vertical coordinate of the selected point. |

| Contour ID | Source Contour          | No. | Derived Feature ID / Name             | Definition and Approval Criteria                                                                                                                                                                                                                                                           |
|------------|-------------------------|-----|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-PS       | Pubic Symphysis Contour | 1   | SPS<br>Superior Pubic Symphysis Point | The most proximal, or superior, point on the pubic symphysis contour. If more than one contour point has the same proximal-most vertical position, select the most medial of the tied points. The principal downstream value of interest is the vertical coordinate of the selected point. |
|            |                         | 2   | IPS<br>Inferior Pubic Symphysis Point | The most distal, or inferior, point on the pubic symphysis contour. If more than one contour point has the same distal-most vertical position, select the most medial of the tied points. The principal downstream value of interest is the vertical coordinate of the selected point.     |

| Contour ID      | Source Contour                                 | No. | Derived Feature ID / Name                 | Definition and Approval Criteria                                                                                                                                                                                                                                   |
|-----------------|------------------------------------------------|-----|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-AC +<br>C-FAC | Acetabular Contour and Acetabular Face Contour | 1   | DAB<br>Distal Acetabular Boundary Point   | The more medial of the two intersection points between the fitted acetabular circle and the fitted acetabular face ellipse. This point must coincide with both fitted contours and defines the medial endpoint of the major axis of the acetabular face ellipse.   |
|                 |                                                | 2   | SAB<br>Superior Acetabular Boundary Point | The more lateral of the two intersection points between the fitted acetabular circle and the fitted acetabular face ellipse. This point must coincide with both fitted contours and defines the lateral endpoint of the major axis of the acetabular face ellipse. |

| <b>Contour ID</b> | <b>Source Contour</b> | <b>No.</b> | <b>Derived Feature ID / Name</b> | <b>Definition and Approval Criteria</b>                                                                                                                                             |
|-------------------|-----------------------|------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-AC              | Acetabular Contour    | 1          | AC Acetabular Centre             | The centre of the circle fitted to the spherical acetabular contour. The generated point should be approved only when it is the geometric centre of the accepted acetabular circle. |

| <b>Contour ID</b> | <b>Source Contour</b> | <b>No.</b> | <b>Derived Feature ID / Name</b> | <b>Definition and Approval Criteria</b>                                                                                                                                       |
|-------------------|-----------------------|------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-FHC             | Femoral Head Contour  | 1          | FHC Femoral Head Centre          | The centre of the circle fitted to the femoral head contour. The generated point should be approved only when it is the geometric centre of the accepted femoral head circle. |

| Contour ID | Source Contour             | No. | Derived Feature ID / Name                | Definition and Approval Criteria                                                                                                                                                     |
|------------|----------------------------|-----|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-GT       | Greater Trochanter Contour | 1   | SGT<br>Superior Greater Trochanter Point | The most superior point on the greater trochanter contour. If more than one contour point has the same superior-most vertical position, select the most lateral of the tied points.  |
|            |                            | 2   | LGT<br>Lateral Greater Trochanter Point  | The most lateral point on the greater trochanter contour. If more than one contour point has the same lateral-most horizontal position, select the most superior of the tied points. |

| Contour ID | Source Contour            | No. | Derived Feature ID / Name               | Definition and Approval Criteria                                                                                                                                                                 |
|------------|---------------------------|-----|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-LT       | Lesser Trochanter Contour | 1   | PLT<br>Proximal Lesser Trochanter Point | The most proximal, or superior, point on the lesser trochanter contour. If more than one contour point has the same proximal-most vertical position, select the most lateral of the tied points. |
|            |                           | 2   | MLT<br>Medial Lesser Trochanter Point   | The most medial point on the lesser trochanter contour. If more than one contour point has the same medial-most horizontal position, select the most superior of the tied points.                |
|            |                           | 3   | DLT<br>Distal Lesser Trochanter Point   | The most distal, or inferior, point on the lesser trochanter contour. If more than one contour point has the same distal-most vertical position, select the most lateral of the tied points.     |

| Contour ID   | Source Contour                                             | No. | Derived Feature ID / Name  | Definition and Approval Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|------------------------------------------------------------|-----|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-DMF + C-GT | Distal Medial Femur Contour and Greater Trochanter Contour | 1   | FA<br>Derived Femoral Axis | Determine the proximal third of the distal medial femur contour using its vertical extent. Restrict the greater trochanter contour to the corresponding vertical range. Use the two resulting contour segments to fit a central line which approximately bisects the medial and lateral boundaries. The resulting line is a candidate proxy for the femoral shaft axis. It should be approved only when it lies centrally between the two contour segments throughout the fitted vertical range and is not visibly biased towards either boundary. |

## 7. Categorical Annotation

In addition to the image annotations defined in Section 3, annotators should make the following categorical annotations regarding image features.

1. For each side (left and right), is the hip native, prosthetic, or other (e.g., tooling is present)?
2. For each annotation, is every associated downstream feature of interest correct, incorrect, or missing?

## A. Annotation ID Summary Table

| Annotation ID | Definition                           |
|---------------|--------------------------------------|
| L-C-ILI       | Left Superior-Lateral Ilium Contour  |
| R-C-ILI       | Right Superior-Lateral Ilium Contour |
| L-C-PO        | Left Pelvic Outlet Contour           |
| R-C-PO        | Right Pelvic Outlet Contour          |
| L-PT          | Left Pelvic Teardrop                 |
| R-PT          | Right Pelvic Teardrop                |
| L-C-IT        | Left Ischial Tuberosity Contour      |
| R-C-IT        | Right Ischial Tuberosity Contour     |
| L-C-PS        | Left Pubic Symphysis Contour         |
| R-C-PS        | Right Pubic Symphysis Contour        |
| L-DSI         | Left Distal Sacroiliac Joint         |
| R-DSI         | Right Distal Sacroiliac Joint        |
| L-C-AC        | Left Acetabular Contour              |
| R-C-AC        | Right Acetabular Contour             |
| L-C-FAC       | Left Acetabular Face Contour         |
| R-C-FAC       | Right Acetabular Face Contour        |
| L-C-FHC       | Left Femoral Head Contour            |
| R-C-FHC       | Right Femoral Head Contour           |
| L-C-GT        | Left Greater Trochanter Contour      |
| R-C-GT        | Right Greater Trochanter Contour     |
| L-C-PMF       | Left Proximal Medial Femur Contour   |
| R-C-PMF       | Right Proximal Medial Femur Contour  |
| L-C-LT        | Left Lesser Trochanter Contour       |
| R-C-LT        | Right Lesser Trochanter Contour      |
| L-C-DMF       | Left Distal Medial Femur Contour     |
| R-C-DMF       | Right Distal Medial Femur Contour    |
| L-FA          | Left Femoral Axis                    |
| R-FA          | Right Femoral Axis                   |
| L-C-OF        | Left Obturator Foramen Contour       |
| R-C-OF        | Right Obturator Foramen Contour      |